Datasets

Day – 10

import pandas as pd

 # 🔧 Step 1: Define prompt-completion pairs

```
data = [
  {
    "prompt": "Write a Python function to add two numbers.",
    "completion": "def add(a, b):\n    return a + b"
  },
  {
    "prompt": "Write a Python function to check if a number is even.",
    "completion": "def is_even(n):\n    return n % 2 == 0"
  },
  {
    "prompt": "Write a Python function to find factorial.",
    "completion": "def factorial(n):\n    return 1 if n == 0 else n * factorial(n - 1)"
  },
  {
    "prompt": "Write a Python function to reverse a string.",
    "completion": "def reverse_string(s):\n    return s[::-1]"
  },
  {
    "prompt": "Write a Python function to find the maximum of three numbers.",
    "completion": "def max_of_three(a, b, c):\n    return max(a, b, c)"
  }
]
```

 # 🔍 Step 2: Convert to DataFrame

df = pd.DataFrame(data)

📄 Step 3: Save as CSV

df.to_csv("code_dataset.csv", index=False)

print("✅ Dataset saved as 'code_dataset.csv'")

🪵 Step 4 (Optional): Save as JSONL (OpenAI fine-tuning format)

with open("code_dataset.jsonl", "w", encoding="utf-8") as f:

  for row in data:

    f.write(f'{{"prompt": "{row["prompt"]}", "completion": "{row["completion"]}"}}\n')

print("✅ Dataset also saved as 'code_dataset.jsonl' (JSON Lines format)")

Here, you've created a Python **list** called data. Each item in the list is a **dictionary** containing two key-value pairs:

- "prompt": The instruction or question you would give to an AI.
- "completion": The ideal answer you want the AI to provide.

This "prompt-completion" format is a standard way to prepare data for training or fine-tuning AI models.

Step 2: Convert to a DataFrame 📊

This part uses the popular pandas library, a powerful tool for working with data tables.

- import pandas as pd: This line imports the library and gives it the common nickname pd.
- pd.DataFrame(data): This command converts your list of dictionaries into a DataFrame, which is essentially a table with rows and columns, making the data easy to work with.

Step 3: Save as a CSV File 📄

This line saves your DataFrame into a file named code_dataset.csv.

- **CSV (Comma-Separated Values)** is a simple spreadsheet format that's easy to open in programs like Excel or Google Sheets.
- index=False is an important argument that prevents pandas from writing an extra, unnecessary row-number column into the file.

Step 4: Save as a JSONL File ✍️

This final step saves the data in **JSONL (JSON Lines)** format.

- **JSONL** is a standard format for AI fine-tuning, especially with platforms like OpenAI. Each line in a .jsonl file is a complete, self-contained JSON object.
- The code opens a file named code_dataset.jsonl and loops through your original data list, writing each prompt-completion pair as a single line in the file.